Updated!
With new US EPA Reports for PM2.5 & O3

Outdoor Air Quality Sensors & Sensor Systems Selection

An Introduction to International Standarization, Testing & Certifications

V2.0 April 21

kunak

You are here

1 **About the trends driving the exponential growth. And challenges for professionals.**

2 A not exhaustive list of Certifications and Testing Protocols and Programs

3 How you too can lead this transformation

kunak

2021 is the year of Suplementary Air Quality Networks

Sensor-based Air monitors are lower in cost, portable and easier to operate than regulatory grade monitors.

They are growing in numbers, widely used to understand air quality conditions.

We are talking about NO, NO2, O3, CO, H2S, SO2, VOCs, CO2, PM10, PM2.5, PM1

#shifthappens

kunak

However, there is currently no regulation for these technologies

These sensors are currently entering the market with limited information as to their repeatability, sensitivity, linearity, cross interferences, temperature impacts, drift, and measurement uncertainty as compared with the more costly instruments typically used by entities such as government agencies.

kunak

Trend #1

Hybrid Networks

Citizens claim for effective actions, but official data is difficult to understand.

Cities look desperately for Smart innovations that provide more spatial and temporal resolution.

In other words
HYPER-LOCAL MONITORING

kunak

Trend #2

Beyond Regulatory

Most developing countries cannot afford traditional technologies.

They need cost-effective monitoring solutions to impose new regulations.

UNEP is doing a terrific job

kunak

Trend #3

This changes Everything

The COVID-19 pandemic defines the need to apply an environmental and sustainable policy based on the reduction of air pollution levels.

kunak

The Utopia

Cities – Industries – Researchers
will make better decissions
based on Suplementary Systems
with larger & accurate data

Cost-effective

Type approval

Accurate

Scalable

Integrable

New Services

kunak

You are here

T

1

Trends driving the exponential growth. Challenges for professionals.

2

A not exhaustive list of Certifications and Testing Protocols and Programs

3

How you too can lead this transformation

kunak

So, how do we select the Best Sensor System?

Well, it depends on the Performance Targets and Testing Protocols

But now it´s the best moment to start considering what certifications and test Programs are out there.

kunak

Meet the family

$5 - $500

Hi, I am a sensor!

We come in different types: Electrochemical, Metaloxide, Laser Scattering.. I need a proper integration to work!

$500- $10,000

Hi, I am a Sensor System, or "Monitor"

I integrate the sensors, electronics, batteries, modems, algorithyms. I am usually designed to be deployed in scalable networks.

$100- $1,000 /year

Hi, I am a Web-based Software

I am here to help: to operate the network, agregate and validate data, calibrate, manage alarms, build reports and convert data into actionable insights.

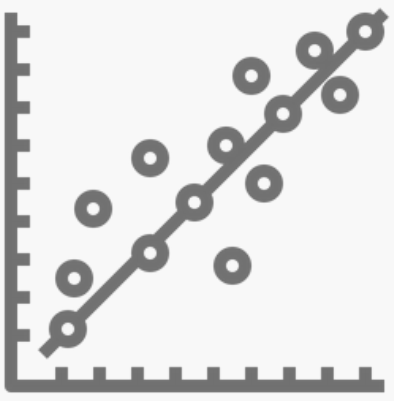
kunak

What are we benchmarking?

The most common metrics are:

R2 (R-Squared)

It represents how well a sensor performs in comparison to the reference instruments.



- R2=1 means the sensor has perfect correlation.
- R2=0 would mean no correlation to a reference monitor.

U(exp) (Expanded Uncertainty)

When using U, we try to demonstrate that the measurements uncertainty of the sensor system is lower than some Data Quality Objectives (DQO) previously stated.
For example, a device with U(exp) < 25% for NO2 would meet the current European Directive requirements for *Indicative Measurements*. This is one of the conditions for Class 1 Certification under future European specification..

Other metrics and factors

R2 and U have their limitations. Many other metrics and statistics are applied:
RMSE, Slope of regression line, Intravariability, Response time, Calibration, Hysteresis, Limit of Detection, Repeatability, Short term drifts, Long term drifts, Cross sensitivities, Effect of temperature and humidity

kunak

Is there Literature?

The few testing centers with enough lab and field equipment are gaining valuable expericence. While guidelines and specifiations are created in Working Groups. But of course there are some papers useful as reference. To cite a few:

- Karagulian, et al., A. Review of the Performance of Low-Cost Sensors for Air Quality Monitoring, 2019

...

- Schneider et al. Toward a Unified Terminology of Processing Levels for Low-Cost Air-Quality Sensors, 2019

- Nuria Castell et al. Can commercial low-cost sensor platforms contribute to air quality monitoring and exposure estimates?, 2017

- Hagan, D. H. and Kroll, J. H.: Assessing the accuracy of low-cost optical particle sensors using a physics-based approach, 2020

- World Meteorological Organization. Lewis et al. Low-cost sensors for the measurement of atmospheric composition: overview of topic and future applications, 2018

kunak

Following, a non exhaustive list of Certification Programs, Testing Facilities, Challenges and Guides to continue educating yourself.

The Category: Non-regulatory Supplemental and Informational Monitoring (NSIM)

kunak

US EPA Reports

New

Air Sensor Toolbox

Recomended Testing Protocols for PM2.5 and Ozone

The EPA has developed two reports outlining recommended testing protocols, metrics, and target values for evaluating the performance of fine particulate matter (PM2.5) and ozone (O3) air sensors. The objective of these reports is to provide a consistent approach for evaluating air sensors specifically for non-regulatory supplemental and informational monitoring (NSIM) applications for use in ambient, outdoor, fixed-site environments. The reports are:

- *Ozone Report*
- *Fine Particulate Matter Report*

https://www.epa.gov/air-sensor-toolbox

kunak

COMING SOON

CEN TC 264 WG 42

EUROPEAN STANDARDS ORGANIZATIONS

Future EU Technical Specification

A protocol describing specific performance requirements and test methods under prescribed laboratory and field conditions for Gas and PMs.

Rules for a new Certification Scheme. The categories are related to Data Quality Objectives of methods set in the Directive 2008/50/EC:

- Class 1 sensor system: consistent with indicative methods
- Class 2 sensor system: consistent with objective estimations methods
- Class 3 sensor system: not formally associated with any mandatory target

https://www.epa.gov/sites/production/files/2019-08/documents/gerboles_european_union_perspective_on_sensor_standards.pdf

kunak

AQSPEC by AQMD

Air Quality Sensor Performance Evaluation Center (AQ-SPEC)

Not a certification. But provides with lots of results under ambient (field) and controlled (laboratory) conditions, made public, for free.

Main Goals & Objectives:
- Evaluate the performance of commercially available "low-cost" air quality sensors in both field and laboratory settings
- Provide guidance and clarity for ever-evolving sensor technology and data interpretation
- Catalyze the successful evolution, development, and use of sensor technology

http://www.aqmd.gov/aq-spec

A NEW TESTING PROTOCOL FOR MOBILE MEASUREMENTS WAS JUST PUBLISHED!

kunak

ASTM WK64899

ASTM INTERNATIONAL
Helping our world work better

New Practice for Performance Evaluation of Ambient Air Quality Sensors and Other Sensor-Based Instruments

Not a Certification. This practice establishes standardized tests and assessment criteria

American Society for Testing and Materials (ASTM) *International is an organization that develops voluntary consensus standards for a variety of products and services.*

https://www.epa.gov/air-sensor-toolbox/air-sensor-performance-targets-and-testing-protocols

kunak

Wildland Fire Challenge

Wildland Fire Sensors Challenge

An evaluation of Sensor systems to measure air pollution during wildland fires

This challenge seeked a field-ready prototype system capable of measuring constituents of smoke, including particulates, carbon monoxide, ozone, and carbon dioxide, over the wide range of levels expected during wildland fires.

Just Published, Atmospheric Environment, 2021:
Matthew S. Landis, et al.
The U.S. EPA wildland fire sensor challenge: Performance and evaluation of solver submitted multi-pollutant sensor systems.

https://www.challenge.gov/challenge/wildland-fire-sensors-challenge/

kunak

INERIS/LNE Certification

A Certification of sensors system for air quality monitoring

A voluntary certification under "AIR Quality Sensor".

- Division A compliant with the data quality objectives (uncertainty, data capture) for Indicative Measurement as described in the Directive 2008/50/EC.
- Division B compliant with the data quality objectives (uncertainty, data capture) for Objective Estimation as described in the Directive 2008/50/EC.
- Division C out of the scope of the Directive 2008/50/EC. Sufficient for citizen science studies, educational action, etc.. defined as Awarness Studies

https://prestations.ineris.fr/en/certification/certification-sensors-system-air-quality-monitoring

kunak

VITO Validation Center

Sensor Validation Center

Voluntary iIndependent determination of the performance of sensors and sensor-based devices.

- *Validation under controlled lab conditions*
- *Validation for industrial applications*
- *Validation for urban applications*
- *Validation for indoor applications*
- *Validation for odour monitoring applications*

https://aqsensors.vito.be/en/sensor-validation-center

Airlab Challenge

Micro-sensors Challenge

The clarity provided by AIRLAB's transparent process enable buyers to choose devices that best fit their needs.

Over a period of four months, Airparif tests the selected devices on 12 pollutants using no less than 28 parameters. Each sensor compete in one or more category of usage (indoor air, outdoor air, static device, mobile device, public awareness...) and is evaluated according to 5 criteria: Accuracy of measurements, ergonomics, relevance of the pollutants measured, portability and cost. Special attention is given to data accessibility. The shortlisted devices are assigned a 1-5 star rating.

http://www.airlab.solutions/en/projects/microsensor-challenge

kunak

The Bluetech Award

CHINA BlueTech AWARDS 中国国际蓝科创新奖

Tackling Air Pollution with Innovative Clean Air Technology

A uniform, systematic assessment methodology to examine all technologies for practical, real-life results

Advanced pollution source and air quality monitoring:

- Ambient and indoor air quality monitoring technologies.
- Pollution source monitoring technologies, such as online VOCs monitoring devices, portable devices.

http://en.bluetechaward.com/

kunak

MCERTs for PMs

IMCERTS Performance Standards for Indicative Ambient Particulate Monitors

The MCERTS certification scheme was established by the UK's Environment Agency and is built around international and European standards to ensure high-quality environmental data monitoring.

The indicative dust monitoring analyser(s) can be operated in one of two ways:
For qualitative measurements: Providing qualitative measurement data for the analysis of particulate pollution trends, and source identification studies based for example on pollution roses etc. Such application can rely on instrument factory calibration only.
For quantitative measurements: Providing measurement data with the uncertainty defined for indicative instruments (±50%), as specified by the Air Quality Directive (2008).

https://www.csagroupuk.org/services/mcerts/

kunak

The Lima Resolution

MUNICIPALIDAD DE LIMA

Guidelines for the performance evaluation of low-cost sensors for Ambient Air Quality Monitoring

A set of metrics and Data Quality Objectives to test sensor & sensor systems, so that the data can be agregated to the analysis, together with data from egulatory networks.

http://smia.munlima.gob.pe/normas/detalle/480

kunak

Making the Invisible Visible

EDF ENVIRONMENTAL DEFENSE FUND™

A guide for mapping hyperlocal air ollution to drive clean air action

Not a Certification Program, not a protocol, but a guide that provides actionable advice on how to design, fund, implement, analyze, and leverage data from a hyperlocal air quality monitoring network, so you can develop solutions to improve health in your community.

PART 1: GETTING STARTED 5

1. Understanding your city's air quality challenges and opportunities 5
2. Exploring tools to measure and map hyperlocal air pollution 11

PART 2: NUTS & BOLTS 13

1. Designing a hyperlocal air monitoring network 13
2. Ready, set, go: Turning on and maintaining your network 28
3. Making sense of the data 40

PART 3: DATA TO ACTION 45

1. Using air pollution data to develop and implement clean air solutions 45
2. Building awareness and encouraging community engagement and support 49
3. Measuring success and maintaining momentum 53

https://www.edf.org/sites/default/files/content/making-the-invisible-visible.pdf

kunak

Assessing air quality through citizen science

European Environment Agency

Engaging citizens in measuring air quality using low-cost devices

The text is partly based on an earlier report prepared by the European Topic Centre (ETC) on Air Pollution and Climate Change Mitigation: ETC/ACM report no 2018/21, Low cost sensor systems for air quality assessment: possibilities and challenges.

https://www.eea.europa.eu/publications/assessing-air-quality-through-citizen-sciencef

https://www.eionet.europa.eu/etcs/etc-atni/products/etc-atni-reports/etc-acm-report-no-2018-21-low-cost-sensor-systems-for-air-quality-assessment-possibilities-and-challenges

kunak

You are here

1 Trends driving the exponential growth. Challenges for professionals.

2 A not exhaustive list of Certifications and Testing Protocols and Programs

3 **How you too can lead this transformation**

kunak

Ok, let´s take a breath

You saw that.

The best minds in the industry are building the blocks for a more reasonable adoption.

Yes. It´s still a bit confusing.

But just take the time to review those reports, and realize that there is consistency in the results of just a handful of manufacturers.

kunak

Your Next Steps

You can lead this movement too, using this knowledge in your projects:

0

Ask for tests results

And don´t buy that device, if it hasn´t gone through plenty of field tests. Do you know how to interpret them?

1

Ask for certifications

Too soon, but stay tuned. Btw, if you think you need them, you are not ready to start this journey yet.

2

Accept 3rd party assesment

Good strategy to benchmark. Which one is a better fit? Do you know how to put that in a tender specification?

3

Design your own protocol

Costly. Do you have the time and resources to develop and later apply?

4

Test, using the protocols

Better. If you have the capacity and time, it´s a smart move.

kunak

The next chapter...?

I think we need to review some real-world Tender Specifications.

And see how the most thoughtful procurement officers are implementing these criteria, today.

Stay Tuned!

As stated, there is no standardised mechanism for quantifying the uncertainty of low cost sensors. Service providers are required to provide evidence of independent testing (either by laboratory or university) using reasonable and commonly accepted methods that the sensors meet the uncertainty requirement for indicative monitoring (±25% for NO_2 and ±50% for $PM_{2.5}$).

Precision is assessed by comparing data from the sensor with reference measurements. This will form the basis for statistical analysis of the data set and calculation of the coefficient of determination (R^2). The coefficient of determination is used as a measure of precision.

Table 10 Required precision

Component	Recommended R^2	Minimum R^2
NO_2	0,8	0,6
O_3	0,8	0,6
SO_2	0,7	0,6
CO	0,8	0,6
PM10	0,7	0,6
PM2.5	0,8	0,6

Evaluation Criteria	Base Package Score	Preferred Options Score
Accuracy		
Typical r^2	4%	4%
Availability of independent test or evaluation reports/formal approval	4%	4%
	2%	2%
Maximum score:	**10%**	**10%**

kunak

Standarization and use of Air Quality Sensors contributes to the SDGs

kunak

Kunak Technologies

Kunak has just presented the new Kunak AIR Pro. An evolution of his predecessors, it is a cost-effective station, designed for type approval. Scalable, accurate and easy to use by professionals and newbies.

Kunak was awarded in the Wildland Fire Challenge

Now, join the movement to improve Air Quality.

Unreasonably easily. Like never before.

kunak

What they say about Kunak

“The three sensors showed an absolute intra-model variability of 0.32 ppb, 0.33 ppb and 0.002 ppm for O3, NO2 and CO measurements, respectively.”

“Sensor nodes have very small form factor and weight that are practical for response field deployments.”

Kunak was awarded by EPA in the Wildland Fire Challenge

“Its very professional design provides a large list of pollutants (NO2, O3, PM1, PM2.5 and PM10) and weather parameters.”

More:

Best option as a multiparametric air quality monitoring system at AQSPEC

www.kunak.es/en/blog/comparative-assessment-aqspec-field-evaluation-reports/

Do you think we can help you manage the Air Quality better?

Get in touch

Miguel Escribano
Business Development Director

mescribano@kunak.es

kunak air

Large data is the goal.
But accurate data is the key.